

Square Values and Square Roots

MA.8.AR.2.3

I can evaluate a square root and find every real solution of $x^2 = p$.

Yesterday you found sides of squares.

Today that gets a symbol — and the symbol is fussy.

Warm-Up

On your own. Four minutes.

Same and different

What is the same about these two, and what is different?

$$\sqrt{36}$$

$$x^2 = 25$$

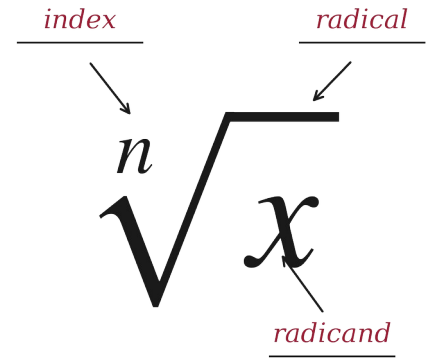
Notes

I. The parts of a radical

A. radical — the symbol $\sqrt{}$ that asks for a root

B. radicand — the number underneath the radical

C. index — the small number in the notch, saying which root to take



Notes

II. The symbol and the equation ask different questions

$\sqrt{25}$ asks for ONE number	$x^2 = 25$ asks for EVERY value
the positive square root	both solutions
$\sqrt{25} = 5$	$x = \pm\sqrt{25} = \pm 5$

Notes

III. The sign lives outside the symbol

- A. **The radical symbol never means two numbers. You write the $\pm$ yourself.**
- B. $\sqrt{25} = 5.$ $-\sqrt{25} = -5.$ $\pm\sqrt{25} = \pm 5.$
- C. Three different expressions, and each one names exactly what it says.

Notes

IV. Solving $x^2 = p$ — three cases

If p is ...	How many solutions	Example
POSITIVE	two, $x = \pm\sqrt{p}$	$x^2 = 49$ gives $x = \pm 7$
ZERO	one, $x = 0$	$x^2 = 0$ gives $x = 0$
NEGATIVE	none that are real	$x^2 = -49$ has no real solution

Example 1

Two questions. Same number underneath.

$$\sqrt{144}$$

$$x^2 = 144$$

Which one has two answers? Answer out loud.

Example 1

Both, worked.

$$\sqrt{144} = 12$$

because $12 \times 12 = 144$

$$x^2 = 144 \quad \text{so} \quad x =$$

$$\pm \sqrt{144} = \pm 12$$

Example 1

Check both.

$$12 \times 12 = 144 \quad \checkmark$$

$$(-12) \times (-12) = 144 \quad \checkmark$$

Example 2

Xavier’s paper. Both answers are on it.

Find the solutions of $b^2 = 25$.	What is the value of $\sqrt{25}$?
$b = 5$ and $b = -5$	5

One of these is wrong. Which?

Example 2

Neither.

Both of Xavier's answers are correct.

Why do they look different?

Example 2

Because they are different questions.

- A. **The equation asks which numbers work. Two of them do.**
- B. **The radical asks for one particular number — the positive one.**
- C. The 25 is the same. The question around it is not.

Whiteboards • Question 1 of 9

Boards up on three.

$$\sqrt{64}$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards · Question 1 of 9

Answer.

$$\sqrt{64}$$

the symbol asks for one number

Answer: 8

Whiteboards • Question 2 of 9

Boards up on three.

$$x^2 = 64$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards • Question 2 of 9

Answer.

$$x^2 = 64$$

which values of x work?

Answer: $x = \pm 8$

Whiteboards • Question 3 of 9

Boards up on three.

$$-\sqrt{64}$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards · Question 3 of 9

Answer.

$$-\sqrt{64}$$

the minus sits outside the radical

Answer: -8

Whiteboards • Question 4 of 9

Boards up on three.

$$x^2 = 0$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards · Question 4 of 9

Answer.

$$x^2 = 0$$

$0 \times 0 = 0$, and nothing else does

Answer: $x = 0$

Whiteboards • Question 5 of 9

Boards up on three.

$$x^2 = -64$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards · Question 5 of 9

Answer.

$$x^2 = -64$$

nothing real squares to a negative

Answer: no real solution

Whiteboards • Question 6 of 9

Boards up on three.

$$\sqrt{-64}$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards • Question 6 of 9

Answer.

$$\sqrt{-64}$$

same refusal, asked with the symbol

Answer: no real value

Whiteboards • Question 7 of 9

Boards up on three.

$$x^2 = 25$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards • Question 7 of 9

Answer.

$$x^2 = 25$$

$$5 \times 5 = 25 \quad \text{and} \quad (-5) \times (-5) = 25$$

Answer: $x = \pm 5$

Whiteboards • Question 8 of 9

Boards up on three.

$$s^2 = \frac{4}{9}$$

Answer it.

A number, or a sentence. Read which it is asking for.

Whiteboards • Question 8 of 9

Answer.

$$s^2 = \frac{4}{9}$$

$$(2/3)^2 = 4/9 \quad \text{and} \quad (-2/3)^2 = 4/9$$

Answer: $s = \pm 2/3$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Jordan says $\sqrt{64} = \pm 8$, because 8 and -8 both square to 64. Is he right?

Write your answer in sentences.

This one is written work. Say which part of what he said is true.

Whiteboards • Question 9 of 9

Answer.

Jordan says $\sqrt{64} = \pm 8$, because 8 and -8 both square to 64. Is he right?

$8^2 = 64$ and $(-8)^2 = 64$ — both are true

Answer: No — $\sqrt{64} = 8$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.